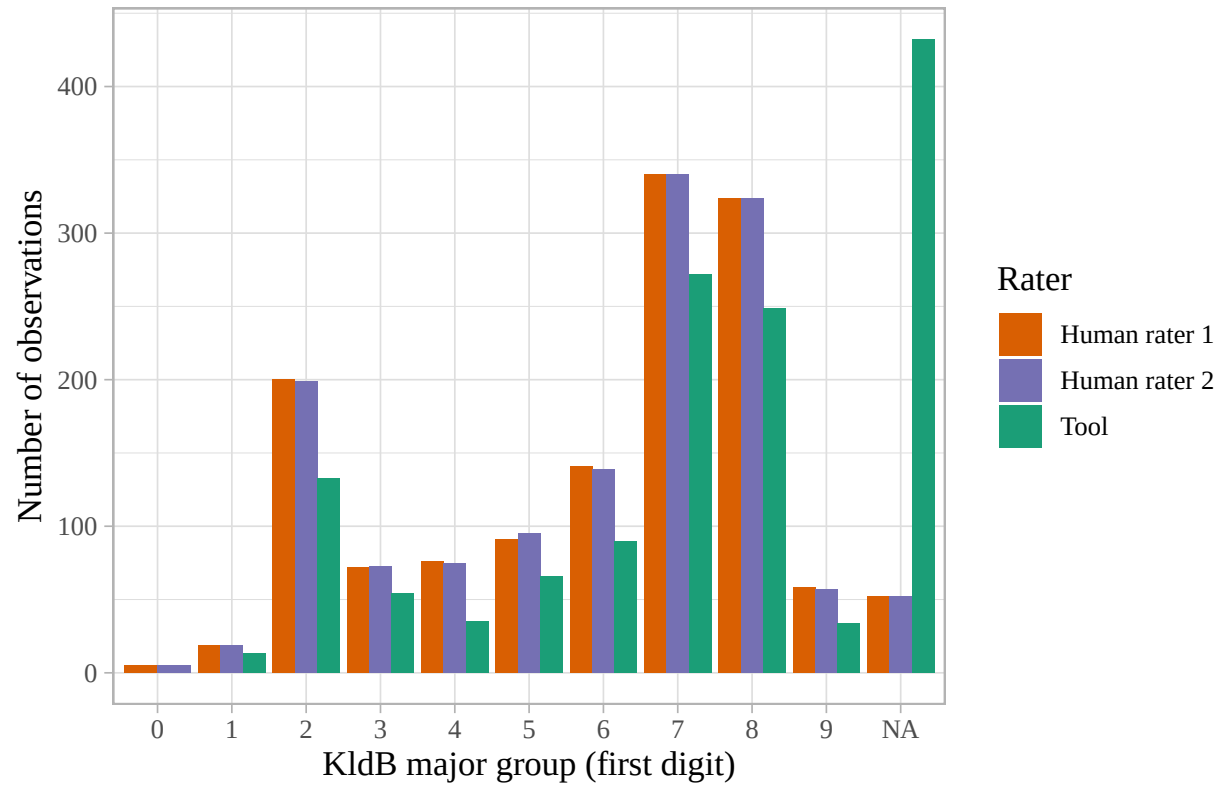
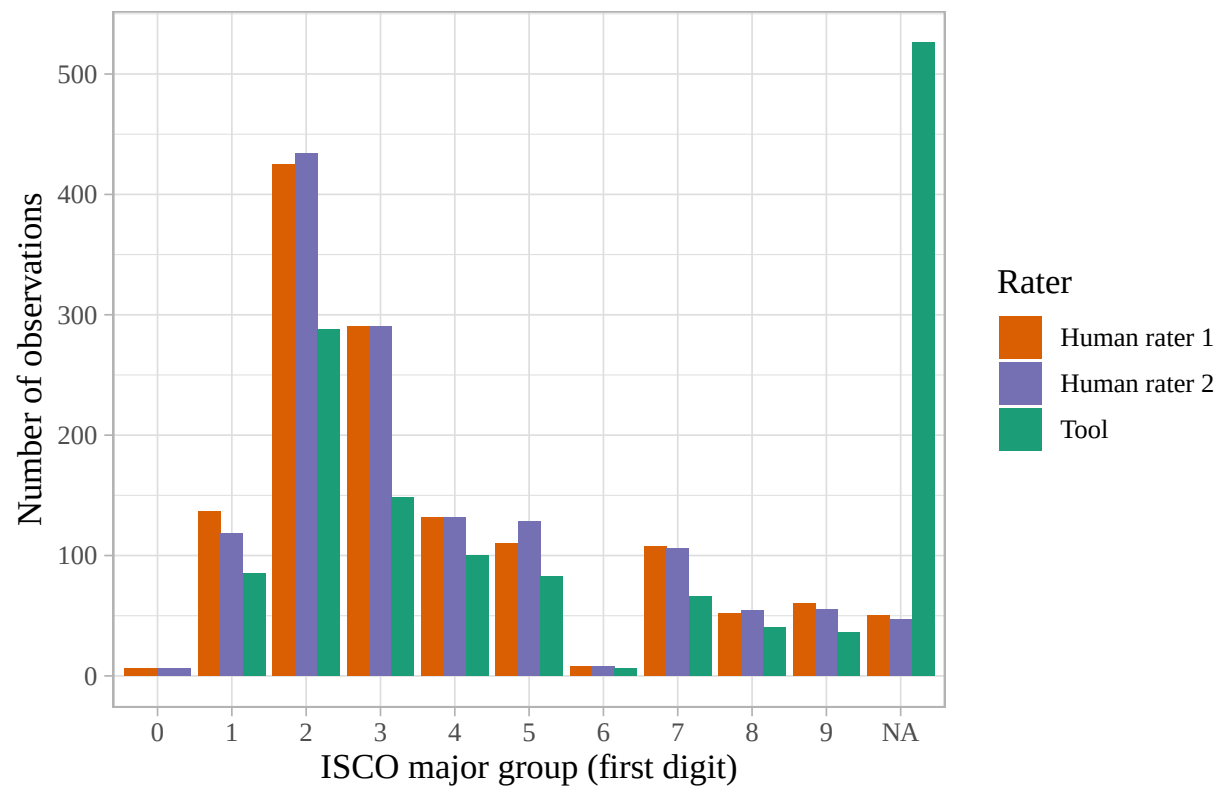


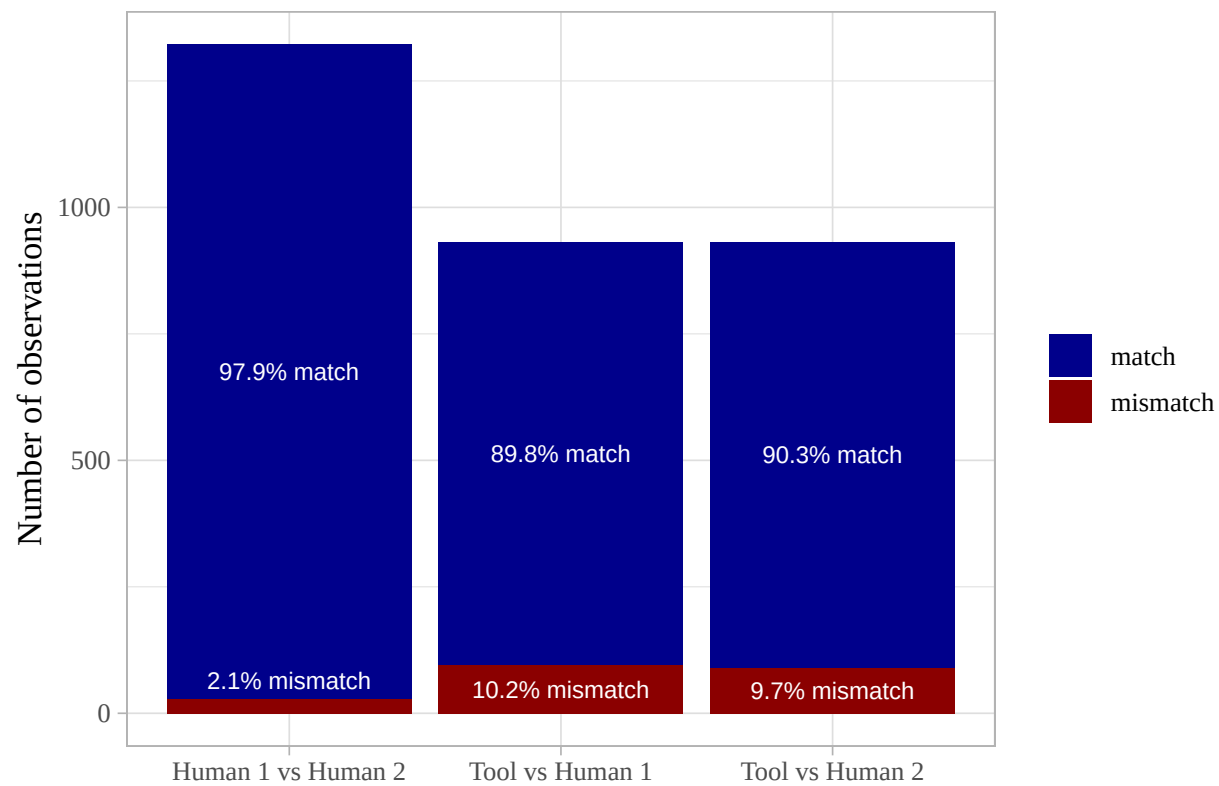
ISCO Alpha Beta Parameter Comparison

```
## [1] "C:/Users/jeroe/Uni_C/noisy_measurement/src/temp_preparation.R"
```

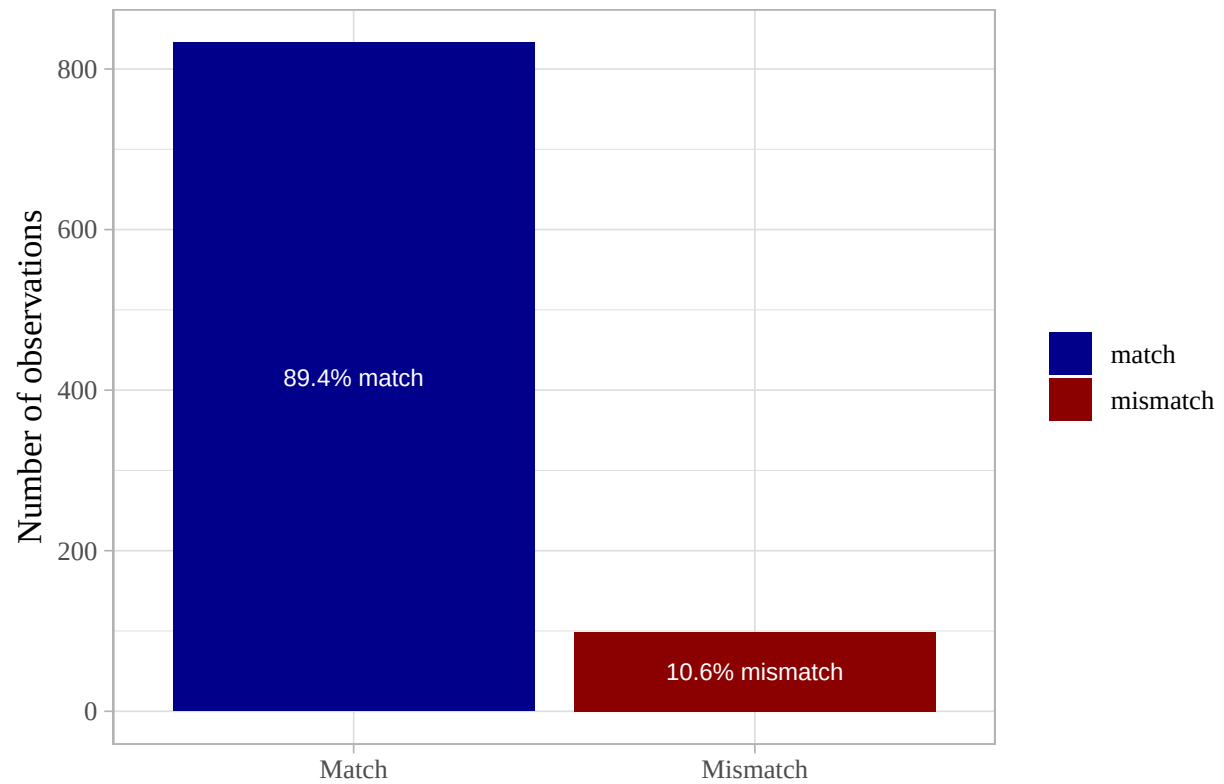




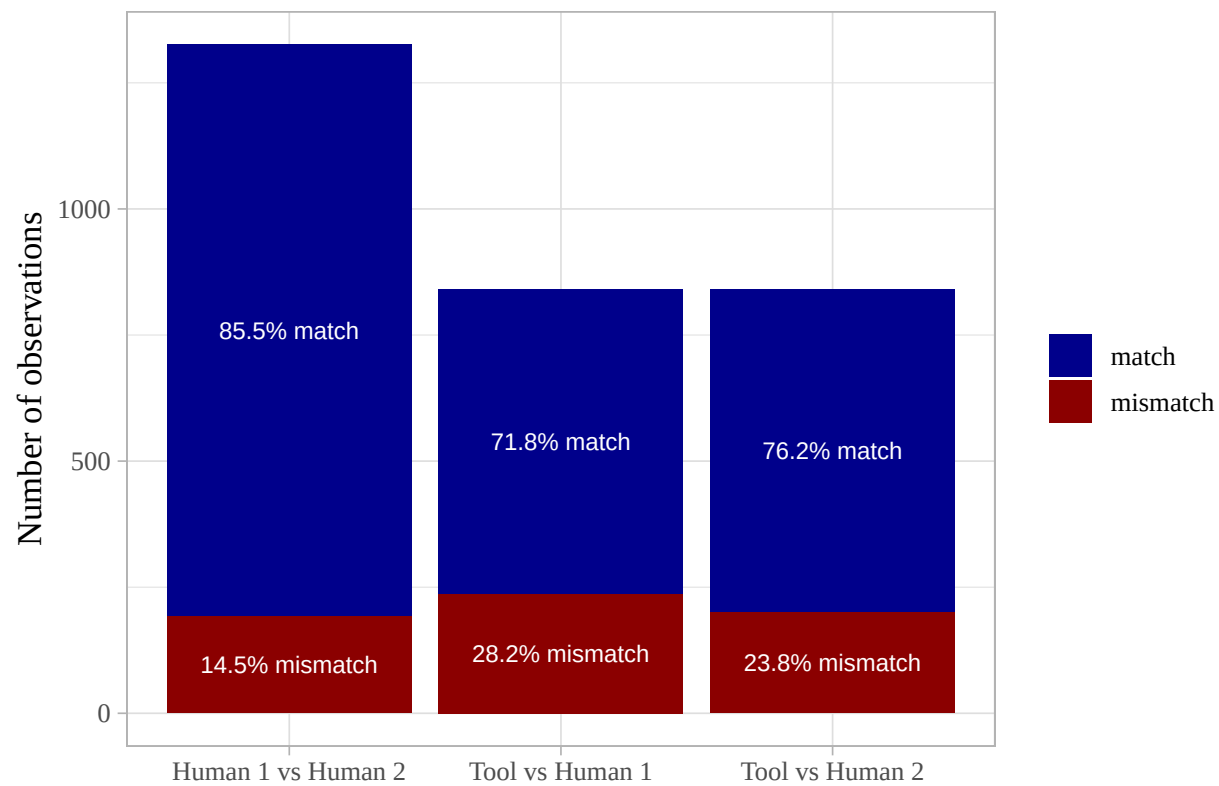
KldB: Pairwise agreement on the first digit



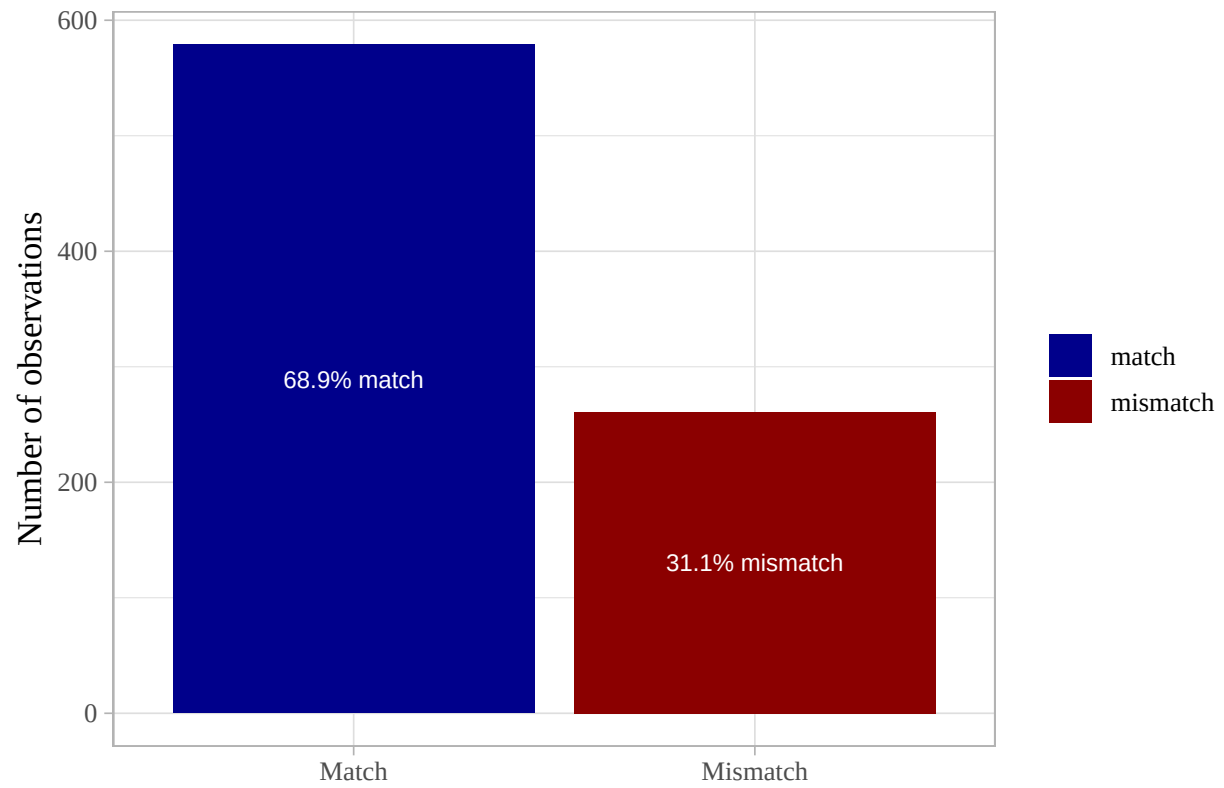
KldB: Agreement across all three sources on the first digit

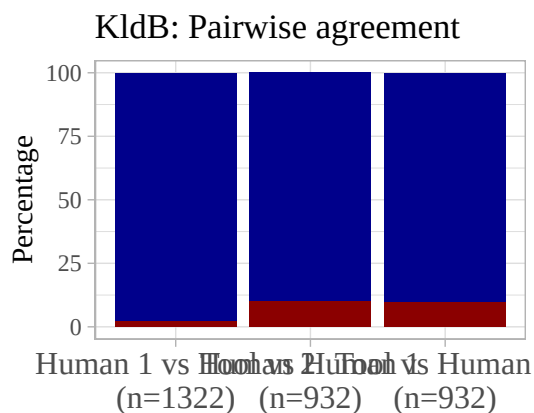
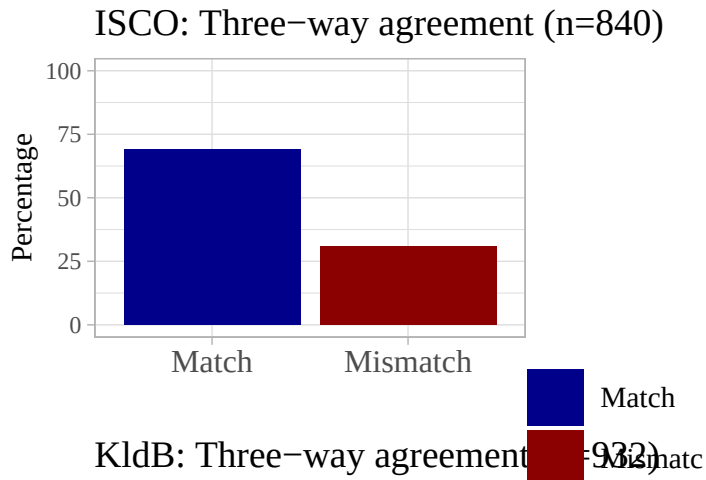
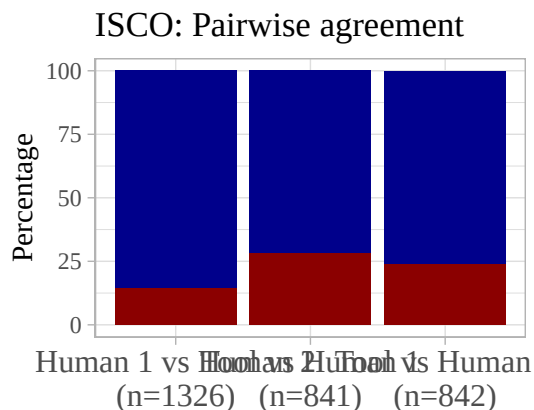


ISCO: Pairwise agreement on the first digit



ISCO: Agreement across all three sources on the first digit





```
##      kldb_tool_vorlaufig kldb_tool_mit_anfniveau res1_kldb res2_kldb isco_tool_vorlaufig isco_tool_mit_
##      <char>                <char>      <char>      <char>                <char>
## 1:      81883              81883      81883      81883              2433
## 2:      52112              52112      52112      52112              8322
## 3:      <NA>               <NA>      83131      83131              <NA>
## 4:      23411              23411      23411      23411              9329
## 5:      72134              72134      72132      72132              2412
## 6:      25104              25104      27224      27224              2144
##      res1_isco res2_isco kldb_infas_num                                kldb_infas_str isco_infas_n
##      <char>      <char>      <char>                                <char>                <char>
## 1:      2433      2433      81883                                [81883] Pharmazie (s.s.T.) - Spezialist      24
## 2:      8322      8322      52112                                [52112] Berufskraftfahrer(Pers./PKW.)-Fachkraft      83
## 3:      3412      3412      83131                                [83131] Heilerziehungspflege, Sonderpaed. -Helfer      34
## 4:      7322      7322      23412                                [23412] Drucktechnik - Fachkraft      73
## 5:      4312      4312      72132                                [72132] Versicherungskaufleute - Fachkraft      43
## 6:      2149      2144      27224                                [27224] Konstruktion und Geraetebau - Experte      21
##
##
## 1: [2433] Akademische und vergleichbare Fachkräfte im Bereich Vertrieb (Technik und Medizin, ohne In
## 2:                                     [8322] Personenkraftwagen-, T
## 3:                                     [3412] Nicht
## 4:
## 5:                                     [4312] Bürokräfte in der St
## 6:
##      kldb_tool_1st res1_kldb_1st res2_kldb_1st isco_tool isco_human1 isco_human2 item_id
##      <char>          <char>          <char>      <char>      <char>      <char>      <int>
```

## 1:	8	8	8	2	2	2	1
## 2:	5	5	5	8	8	8	2
## 3:	<NA>	8	8	<NA>	3	3	3
## 4:	2	2	2	9	7	7	4
## 5:	7	7	7	2	4	4	5
## 6:	2	2	2	2	2	2	6

[1] 235

[1] 80

[1] 42

[1] 0

Items after filtering: 357

Observations: 1071

Classes: 2

Class levels: 2, 3

##

=== Fitting model: alpha11_beta1 ===

Running MCMC with 4 parallel chains, with 1 thread(s) per chain...

##

```
## Chain 1 Iteration: 1 / 5000 [ 0%] (Warmup)
## Chain 2 Iteration: 1 / 5000 [ 0%] (Warmup)
## Chain 3 Iteration: 1 / 5000 [ 0%] (Warmup)
## Chain 4 Iteration: 1 / 5000 [ 0%] (Warmup)
## Chain 1 Iteration: 200 / 5000 [ 4%] (Warmup)
## Chain 2 Iteration: 200 / 5000 [ 4%] (Warmup)
## Chain 3 Iteration: 200 / 5000 [ 4%] (Warmup)
## Chain 1 Iteration: 400 / 5000 [ 8%] (Warmup)
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## Chain 2 Iteration: 600 / 5000 [ 12%] (Warmup)
## Chain 1 Iteration: 800 / 5000 [ 16%] (Warmup)
## Chain 4 Iteration: 600 / 5000 [ 12%] (Warmup)
## Chain 3 Iteration: 800 / 5000 [ 16%] (Warmup)
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## Chain 4 Iteration: 800 / 5000 [ 16%] (Warmup)
## Chain 1 Iteration: 1000 / 5000 [ 20%] (Warmup)
## Chain 1 Iteration: 1001 / 5000 [ 20%] (Sampling)
## Chain 3 Iteration: 1000 / 5000 [ 20%] (Warmup)
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## Chain 1 Iteration: 1200 / 5000 [ 24%] (Sampling)
## Chain 4 Iteration: 1000 / 5000 [ 20%] (Warmup)
## Chain 4 Iteration: 1001 / 5000 [ 20%] (Sampling)
## Chain 3 Iteration: 1200 / 5000 [ 24%] (Sampling)
## Chain 1 Iteration: 1400 / 5000 [ 28%] (Sampling)
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## Chain 2 Iteration: 1200 / 5000 [ 24%] (Sampling)
## Chain 4 Iteration: 1200 / 5000 [ 24%] (Sampling)
## Chain 1 Iteration: 1600 / 5000 [ 32%] (Sampling)
## Chain 3 Iteration: 1400 / 5000 [ 28%] (Sampling)
## Chain 2 Iteration: 1400 / 5000 [ 28%] (Sampling)
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## Chain 4 Iteration: 1400 / 5000 [ 28%] (Sampling)
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## Chain 1 Iteration: 2000 / 5000 [ 40%] (Sampling)
## Chain 4 Iteration: 1600 / 5000 [ 32%] (Sampling)
## Chain 2 Iteration: 1800 / 5000 [ 36%] (Sampling)
## Chain 3 Iteration: 1800 / 5000 [ 36%] (Sampling)
## Chain 1 Iteration: 2200 / 5000 [ 44%] (Sampling)
## Chain 3 Iteration: 2000 / 5000 [ 40%] (Sampling)
## Chain 2 Iteration: 2000 / 5000 [ 40%] (Sampling)
## Chain 4 Iteration: 1800 / 5000 [ 36%] (Sampling)
## Chain 1 Iteration: 2400 / 5000 [ 48%] (Sampling)
## Chain 3 Iteration: 2200 / 5000 [ 44%] (Sampling)
## Chain 1 Iteration: 2600 / 5000 [ 52%] (Sampling)
## Chain 2 Iteration: 2200 / 5000 [ 44%] (Sampling)
## Chain 4 Iteration: 2000 / 5000 [ 40%] (Sampling)
## Chain 1 Iteration: 2800 / 5000 [ 56%] (Sampling)
## Chain 3 Iteration: 2400 / 5000 [ 48%] (Sampling)
## Chain 2 Iteration: 2400 / 5000 [ 48%] (Sampling)
## Chain 4 Iteration: 2200 / 5000 [ 44%] (Sampling)
## Chain 1 Iteration: 3000 / 5000 [ 60%] (Sampling)
## Chain 3 Iteration: 2600 / 5000 [ 52%] (Sampling)
## Chain 2 Iteration: 2600 / 5000 [ 52%] (Sampling)
## Chain 1 Iteration: 3200 / 5000 [ 64%] (Sampling)
## Chain 4 Iteration: 2400 / 5000 [ 48%] (Sampling)
## Chain 3 Iteration: 2800 / 5000 [ 56%] (Sampling)
## Chain 2 Iteration: 2800 / 5000 [ 56%] (Sampling)
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## Chain 4 Iteration: 2600 / 5000 [ 52%] (Sampling)
## Chain 2 Iteration: 3000 / 5000 [ 60%] (Sampling)
## Chain 3 Iteration: 3000 / 5000 [ 60%] (Sampling)
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## Chain 1 Iteration: 3800 / 5000 [ 76%] (Sampling)
## Chain 2 Iteration: 3200 / 5000 [ 64%] (Sampling)
## Chain 3 Iteration: 3200 / 5000 [ 64%] (Sampling)
## Chain 4 Iteration: 2800 / 5000 [ 56%] (Sampling)
## Chain 1 Iteration: 4000 / 5000 [ 80%] (Sampling)
## Chain 2 Iteration: 3400 / 5000 [ 68%] (Sampling)
## Chain 3 Iteration: 3400 / 5000 [ 68%] (Sampling)
## Chain 4 Iteration: 3000 / 5000 [ 60%] (Sampling)
## Chain 1 Iteration: 4200 / 5000 [ 84%] (Sampling)
## Chain 2 Iteration: 3600 / 5000 [ 72%] (Sampling)
## Chain 3 Iteration: 3600 / 5000 [ 72%] (Sampling)
## Chain 1 Iteration: 4400 / 5000 [ 88%] (Sampling)
## Chain 4 Iteration: 3200 / 5000 [ 64%] (Sampling)
## Chain 2 Iteration: 3800 / 5000 [ 76%] (Sampling)
## Chain 3 Iteration: 3800 / 5000 [ 76%] (Sampling)
## Chain 1 Iteration: 4600 / 5000 [ 92%] (Sampling)

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## Chain 4 Iteration: 3400 / 5000 [ 68%] (Sampling)
## Chain 3 Iteration: 4000 / 5000 [ 80%] (Sampling)
## Chain 1 Iteration: 4800 / 5000 [ 96%] (Sampling)
## Chain 2 Iteration: 4000 / 5000 [ 80%] (Sampling)
## Chain 1 Iteration: 5000 / 5000 [100%] (Sampling)
## Chain 1 finished in 43.1 seconds.
## Chain 4 Iteration: 3600 / 5000 [ 72%] (Sampling)
## Chain 2 Iteration: 4200 / 5000 [ 84%] (Sampling)
## Chain 3 Iteration: 4200 / 5000 [ 84%] (Sampling)
## Chain 3 Iteration: 4400 / 5000 [ 88%] (Sampling)
## Chain 2 Iteration: 4400 / 5000 [ 88%] (Sampling)
## Chain 4 Iteration: 3800 / 5000 [ 76%] (Sampling)
## Chain 2 Iteration: 4600 / 5000 [ 92%] (Sampling)
## Chain 3 Iteration: 4600 / 5000 [ 92%] (Sampling)
## Chain 4 Iteration: 4000 / 5000 [ 80%] (Sampling)
## Chain 2 Iteration: 4800 / 5000 [ 96%] (Sampling)
## Chain 3 Iteration: 4800 / 5000 [ 96%] (Sampling)
## Chain 4 Iteration: 4200 / 5000 [ 84%] (Sampling)
## Chain 2 Iteration: 5000 / 5000 [100%] (Sampling)
## Chain 2 finished in 51.9 seconds.
## Chain 3 Iteration: 5000 / 5000 [100%] (Sampling)
## Chain 3 finished in 52.1 seconds.
## Chain 4 Iteration: 4400 / 5000 [ 88%] (Sampling)
## Chain 4 Iteration: 4600 / 5000 [ 92%] (Sampling)
## Chain 4 Iteration: 4800 / 5000 [ 96%] (Sampling)
## Chain 4 Iteration: 5000 / 5000 [100%] (Sampling)
## Chain 4 finished in 61.3 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 52.1 seconds.
## Total execution time: 61.4 seconds.
##
## No R-hat values above 1.1
##
## === Fitting model: alpha11_beta5diag ===
## Running MCMC with 4 parallel chains, with 1 thread(s) per chain...
##
## Chain 1 Iteration:    1 / 5000 [  0%] (Warmup)
## Chain 2 Iteration:    1 / 5000 [  0%] (Warmup)
## Chain 3 Iteration:    1 / 5000 [  0%] (Warmup)
## Chain 4 Iteration:    1 / 5000 [  0%] (Warmup)
## Chain 1 Iteration:  200 / 5000 [  4%] (Warmup)
## Chain 3 Iteration:  200 / 5000 [  4%] (Warmup)
## Chain 2 Iteration:  200 / 5000 [  4%] (Warmup)
## Chain 4 Iteration:  200 / 5000 [  4%] (Warmup)
## Chain 1 Iteration:  400 / 5000 [  8%] (Warmup)
## Chain 3 Iteration:  400 / 5000 [  8%] (Warmup)
## Chain 4 Iteration:  400 / 5000 [  8%] (Warmup)
## Chain 2 Iteration:  400 / 5000 [  8%] (Warmup)
## Chain 1 Iteration:  600 / 5000 [ 12%] (Warmup)
## Chain 4 Iteration:  600 / 5000 [ 12%] (Warmup)
## Chain 1 Iteration:  800 / 5000 [ 16%] (Warmup)
## Chain 2 Iteration:  600 / 5000 [ 12%] (Warmup)
## Chain 3 Iteration:  600 / 5000 [ 12%] (Warmup)

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## Chain 1 Iteration: 1000 / 5000 [ 20%] (Warmup)
## Chain 4 Iteration: 800 / 5000 [ 16%] (Warmup)
## Chain 1 Iteration: 1001 / 5000 [ 20%] (Sampling)
## Chain 2 Iteration: 800 / 5000 [ 16%] (Warmup)
## Chain 3 Iteration: 800 / 5000 [ 16%] (Warmup)
## Chain 1 Iteration: 1200 / 5000 [ 24%] (Sampling)
## Chain 4 Iteration: 1000 / 5000 [ 20%] (Warmup)
## Chain 4 Iteration: 1001 / 5000 [ 20%] (Sampling)
## Chain 2 Iteration: 1000 / 5000 [ 20%] (Warmup)
## Chain 3 Iteration: 1000 / 5000 [ 20%] (Warmup)
## Chain 2 Iteration: 1001 / 5000 [ 20%] (Sampling)
## Chain 3 Iteration: 1001 / 5000 [ 20%] (Sampling)
## Chain 1 Iteration: 1400 / 5000 [ 28%] (Sampling)
## Chain 4 Iteration: 1200 / 5000 [ 24%] (Sampling)
## Chain 3 Iteration: 1200 / 5000 [ 24%] (Sampling)
## Chain 2 Iteration: 1200 / 5000 [ 24%] (Sampling)
## Chain 1 Iteration: 1600 / 5000 [ 32%] (Sampling)
## Chain 4 Iteration: 1400 / 5000 [ 28%] (Sampling)
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## Chain 4 Iteration: 1600 / 5000 [ 32%] (Sampling)
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## Chain 3 Iteration: 1600 / 5000 [ 32%] (Sampling)
## Chain 1 Iteration: 2000 / 5000 [ 40%] (Sampling)
## Chain 4 Iteration: 1800 / 5000 [ 36%] (Sampling)
## Chain 1 Iteration: 2200 / 5000 [ 44%] (Sampling)
## Chain 2 Iteration: 1600 / 5000 [ 32%] (Sampling)
## Chain 3 Iteration: 1800 / 5000 [ 36%] (Sampling)
## Chain 4 Iteration: 2000 / 5000 [ 40%] (Sampling)
## Chain 1 Iteration: 2400 / 5000 [ 48%] (Sampling)
## Chain 4 Iteration: 2200 / 5000 [ 44%] (Sampling)
## Chain 3 Iteration: 2000 / 5000 [ 40%] (Sampling)
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## Chain 1 Iteration: 2600 / 5000 [ 52%] (Sampling)
## Chain 4 Iteration: 2400 / 5000 [ 48%] (Sampling)
## Chain 3 Iteration: 2200 / 5000 [ 44%] (Sampling)
## Chain 1 Iteration: 2800 / 5000 [ 56%] (Sampling)
## Chain 2 Iteration: 2000 / 5000 [ 40%] (Sampling)
## Chain 4 Iteration: 2600 / 5000 [ 52%] (Sampling)
## Chain 3 Iteration: 2400 / 5000 [ 48%] (Sampling)
## Chain 1 Iteration: 3000 / 5000 [ 60%] (Sampling)
## Chain 4 Iteration: 2800 / 5000 [ 56%] (Sampling)
## Chain 2 Iteration: 2200 / 5000 [ 44%] (Sampling)
## Chain 3 Iteration: 2600 / 5000 [ 52%] (Sampling)
## Chain 1 Iteration: 3200 / 5000 [ 64%] (Sampling)
## Chain 4 Iteration: 3000 / 5000 [ 60%] (Sampling)
## Chain 1 Iteration: 3400 / 5000 [ 68%] (Sampling)
## Chain 3 Iteration: 2800 / 5000 [ 56%] (Sampling)
## Chain 2 Iteration: 2400 / 5000 [ 48%] (Sampling)
## Chain 4 Iteration: 3200 / 5000 [ 64%] (Sampling)
## Chain 1 Iteration: 3600 / 5000 [ 72%] (Sampling)
## Chain 4 Iteration: 3400 / 5000 [ 68%] (Sampling)
## Chain 3 Iteration: 3000 / 5000 [ 60%] (Sampling)
## Chain 2 Iteration: 2600 / 5000 [ 52%] (Sampling)

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## Chain 1 Iteration: 3800 / 5000 [ 76%] (Sampling)
## Chain 4 Iteration: 3600 / 5000 [ 72%] (Sampling)
## Chain 3 Iteration: 3200 / 5000 [ 64%] (Sampling)
## Chain 1 Iteration: 4000 / 5000 [ 80%] (Sampling)
## Chain 2 Iteration: 2800 / 5000 [ 56%] (Sampling)
## Chain 4 Iteration: 3800 / 5000 [ 76%] (Sampling)
## Chain 1 Iteration: 4200 / 5000 [ 84%] (Sampling)
## Chain 3 Iteration: 3400 / 5000 [ 68%] (Sampling)
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## Chain 1 Iteration: 4400 / 5000 [ 88%] (Sampling)
## Chain 2 Iteration: 3000 / 5000 [ 60%] (Sampling)
## Chain 3 Iteration: 3600 / 5000 [ 72%] (Sampling)
## Chain 4 Iteration: 4200 / 5000 [ 84%] (Sampling)
## Chain 1 Iteration: 4600 / 5000 [ 92%] (Sampling)
## Chain 3 Iteration: 3800 / 5000 [ 76%] (Sampling)
## Chain 4 Iteration: 4400 / 5000 [ 88%] (Sampling)
## Chain 2 Iteration: 3200 / 5000 [ 64%] (Sampling)
## Chain 1 Iteration: 4800 / 5000 [ 96%] (Sampling)
## Chain 3 Iteration: 4000 / 5000 [ 80%] (Sampling)
## Chain 4 Iteration: 4600 / 5000 [ 92%] (Sampling)
## Chain 1 Iteration: 5000 / 5000 [100%] (Sampling)
## Chain 1 finished in 38.4 seconds.
## Chain 2 Iteration: 3400 / 5000 [ 68%] (Sampling)
## Chain 3 Iteration: 4200 / 5000 [ 84%] (Sampling)
## Chain 4 Iteration: 4800 / 5000 [ 96%] (Sampling)
## Chain 4 Iteration: 5000 / 5000 [100%] (Sampling)
## Chain 4 finished in 41.2 seconds.
## Chain 2 Iteration: 3600 / 5000 [ 72%] (Sampling)
## Chain 3 Iteration: 4400 / 5000 [ 88%] (Sampling)
## Chain 3 Iteration: 4600 / 5000 [ 92%] (Sampling)
## Chain 2 Iteration: 3800 / 5000 [ 76%] (Sampling)
## Chain 3 Iteration: 4800 / 5000 [ 96%] (Sampling)
## Chain 2 Iteration: 4000 / 5000 [ 80%] (Sampling)
## Chain 3 Iteration: 5000 / 5000 [100%] (Sampling)
## Chain 3 finished in 46.9 seconds.
## Chain 2 Iteration: 4200 / 5000 [ 84%] (Sampling)
## Chain 2 Iteration: 4400 / 5000 [ 88%] (Sampling)
## Chain 2 Iteration: 4600 / 5000 [ 92%] (Sampling)
## Chain 2 Iteration: 4800 / 5000 [ 96%] (Sampling)
## Chain 2 Iteration: 5000 / 5000 [100%] (Sampling)
## Chain 2 finished in 58.1 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 46.2 seconds.
## Total execution time: 58.2 seconds.
##
## No R-hat values above 1.1
##
## === Fitting model: alpha1510_beta1 ===
## Running MCMC with 4 parallel chains, with 1 thread(s) per chain...
##
## Chain 1 Iteration:      1 / 5000 [  0%] (Warmup)
## Chain 2 Iteration:      1 / 5000 [  0%] (Warmup)
## Chain 3 Iteration:      1 / 5000 [  0%] (Warmup)

```

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## Chain 4 Iteration:      1 / 5000 [ 0%] (Warmup)
## Chain 1 Iteration:    200 / 5000 [ 4%] (Warmup)
## Chain 3 Iteration:    200 / 5000 [ 4%] (Warmup)
## Chain 4 Iteration:    200 / 5000 [ 4%] (Warmup)
## Chain 2 Iteration:    200 / 5000 [ 4%] (Warmup)
## Chain 1 Iteration:    400 / 5000 [ 8%] (Warmup)
## Chain 3 Iteration:    400 / 5000 [ 8%] (Warmup)
## Chain 2 Iteration:    400 / 5000 [ 8%] (Warmup)
## Chain 4 Iteration:    400 / 5000 [ 8%] (Warmup)
## Chain 1 Iteration:    600 / 5000 [12%] (Warmup)
## Chain 3 Iteration:    600 / 5000 [12%] (Warmup)
## Chain 1 Iteration:    800 / 5000 [16%] (Warmup)
## Chain 2 Iteration:    600 / 5000 [12%] (Warmup)
## Chain 4 Iteration:    600 / 5000 [12%] (Warmup)
## Chain 3 Iteration:    800 / 5000 [16%] (Warmup)
## Chain 2 Iteration:    800 / 5000 [16%] (Warmup)
## Chain 4 Iteration:    800 / 5000 [16%] (Warmup)
## Chain 1 Iteration:   1000 / 5000 [20%] (Warmup)
## Chain 1 Iteration:   1001 / 5000 [20%] (Sampling)
## Chain 2 Iteration:   1000 / 5000 [20%] (Warmup)
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## Chain 2 Iteration:   1001 / 5000 [20%] (Sampling)
## Chain 3 Iteration:   1001 / 5000 [20%] (Sampling)
## Chain 4 Iteration:   1000 / 5000 [20%] (Warmup)
## Chain 4 Iteration:   1001 / 5000 [20%] (Sampling)
## Chain 1 Iteration:   1200 / 5000 [24%] (Sampling)
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## Chain 1 Iteration:   2000 / 5000 [40%] (Sampling)
## Chain 4 Iteration:   2200 / 5000 [44%] (Sampling)
## Chain 2 Iteration:   2400 / 5000 [48%] (Sampling)
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## Chain 3 Iteration:   2000 / 5000 [40%] (Sampling)
## Chain 1 Iteration:   2200 / 5000 [44%] (Sampling)
## Chain 2 Iteration:   2600 / 5000 [52%] (Sampling)
## Chain 4 Iteration:   2600 / 5000 [52%] (Sampling)
## Chain 3 Iteration:   2200 / 5000 [44%] (Sampling)
## Chain 1 Iteration:   2400 / 5000 [48%] (Sampling)

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## Chain 2 Iteration: 2800 / 5000 [ 56%] (Sampling)
## Chain 4 Iteration: 2800 / 5000 [ 56%] (Sampling)
## Chain 2 Iteration: 3000 / 5000 [ 60%] (Sampling)
## Chain 1 Iteration: 2600 / 5000 [ 52%] (Sampling)
## Chain 3 Iteration: 2400 / 5000 [ 48%] (Sampling)
## Chain 4 Iteration: 3000 / 5000 [ 60%] (Sampling)
## Chain 2 Iteration: 3200 / 5000 [ 64%] (Sampling)
## Chain 4 Iteration: 3200 / 5000 [ 64%] (Sampling)
## Chain 1 Iteration: 2800 / 5000 [ 56%] (Sampling)
## Chain 3 Iteration: 2600 / 5000 [ 52%] (Sampling)
## Chain 2 Iteration: 3400 / 5000 [ 68%] (Sampling)
## Chain 4 Iteration: 3400 / 5000 [ 68%] (Sampling)
## Chain 1 Iteration: 3000 / 5000 [ 60%] (Sampling)
## Chain 3 Iteration: 2800 / 5000 [ 56%] (Sampling)
## Chain 2 Iteration: 3600 / 5000 [ 72%] (Sampling)
## Chain 4 Iteration: 3600 / 5000 [ 72%] (Sampling)
## Chain 2 Iteration: 3800 / 5000 [ 76%] (Sampling)
## Chain 4 Iteration: 3800 / 5000 [ 76%] (Sampling)
## Chain 1 Iteration: 3200 / 5000 [ 64%] (Sampling)
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## Chain 2 Iteration: 4000 / 5000 [ 80%] (Sampling)
## Chain 4 Iteration: 4000 / 5000 [ 80%] (Sampling)
## Chain 1 Iteration: 3400 / 5000 [ 68%] (Sampling)
## Chain 3 Iteration: 3200 / 5000 [ 64%] (Sampling)
## Chain 2 Iteration: 4200 / 5000 [ 84%] (Sampling)
## Chain 4 Iteration: 4200 / 5000 [ 84%] (Sampling)
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## Chain 1 Iteration: 3800 / 5000 [ 76%] (Sampling)
## Chain 3 Iteration: 3600 / 5000 [ 72%] (Sampling)
## Chain 2 Iteration: 4800 / 5000 [ 96%] (Sampling)
## Chain 4 Iteration: 4800 / 5000 [ 96%] (Sampling)
## Chain 1 Iteration: 4000 / 5000 [ 80%] (Sampling)
## Chain 2 Iteration: 5000 / 5000 [100%] (Sampling)
## Chain 4 Iteration: 5000 / 5000 [100%] (Sampling)
## Chain 2 finished in 38.8 seconds.
## Chain 4 finished in 38.8 seconds.
## Chain 3 Iteration: 3800 / 5000 [ 76%] (Sampling)
## Chain 1 Iteration: 4200 / 5000 [ 84%] (Sampling)
## Chain 3 Iteration: 4000 / 5000 [ 80%] (Sampling)
## Chain 1 Iteration: 4400 / 5000 [ 88%] (Sampling)
## Chain 3 Iteration: 4200 / 5000 [ 84%] (Sampling)
## Chain 1 Iteration: 4600 / 5000 [ 92%] (Sampling)
## Chain 3 Iteration: 4400 / 5000 [ 88%] (Sampling)
## Chain 1 Iteration: 4800 / 5000 [ 96%] (Sampling)
## Chain 3 Iteration: 4600 / 5000 [ 92%] (Sampling)
## Chain 1 Iteration: 5000 / 5000 [100%] (Sampling)
## Chain 1 finished in 48.0 seconds.
## Chain 3 Iteration: 4800 / 5000 [ 96%] (Sampling)
## Chain 3 Iteration: 5000 / 5000 [100%] (Sampling)

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## Chain 3 finished in 51.4 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 44.3 seconds.
## Total execution time: 51.5 seconds.
##
## No R-hat values above 1.1
##
## === Fitting model: alpha1510_beta5diag ===
## Running MCMC with 4 parallel chains, with 1 thread(s) per chain...
##
## Chain 1 Iteration:    1 / 5000 [ 0%] (Warmup)
## Chain 2 Iteration:    1 / 5000 [ 0%] (Warmup)
## Chain 3 Iteration:    1 / 5000 [ 0%] (Warmup)
## Chain 4 Iteration:    1 / 5000 [ 0%] (Warmup)
## Chain 1 Iteration:   200 / 5000 [ 4%] (Warmup)
## Chain 3 Iteration:   200 / 5000 [ 4%] (Warmup)
## Chain 4 Iteration:   200 / 5000 [ 4%] (Warmup)
## Chain 1 Iteration:   400 / 5000 [ 8%] (Warmup)
## Chain 2 Iteration:   200 / 5000 [ 4%] (Warmup)
## Chain 3 Iteration:   400 / 5000 [ 8%] (Warmup)
## Chain 4 Iteration:   400 / 5000 [ 8%] (Warmup)
## Chain 1 Iteration:   600 / 5000 [12%] (Warmup)
## Chain 2 Iteration:   400 / 5000 [ 8%] (Warmup)
## Chain 3 Iteration:   600 / 5000 [12%] (Warmup)
## Chain 1 Iteration:   800 / 5000 [16%] (Warmup)
## Chain 4 Iteration:   600 / 5000 [12%] (Warmup)
## Chain 2 Iteration:   600 / 5000 [12%] (Warmup)
## Chain 3 Iteration:   800 / 5000 [16%] (Warmup)
## Chain 4 Iteration:   800 / 5000 [16%] (Warmup)
## Chain 1 Iteration:  1000 / 5000 [20%] (Warmup)
## Chain 1 Iteration:  1001 / 5000 [20%] (Sampling)
## Chain 3 Iteration:  1000 / 5000 [20%] (Warmup)
## Chain 2 Iteration:   800 / 5000 [16%] (Warmup)
## Chain 3 Iteration:  1001 / 5000 [20%] (Sampling)
## Chain 4 Iteration:  1000 / 5000 [20%] (Warmup)
## Chain 4 Iteration:  1001 / 5000 [20%] (Sampling)
## Chain 3 Iteration:  1200 / 5000 [24%] (Sampling)
## Chain 1 Iteration:  1200 / 5000 [24%] (Sampling)
## Chain 2 Iteration:  1000 / 5000 [20%] (Warmup)
## Chain 2 Iteration:  1001 / 5000 [20%] (Sampling)
## Chain 4 Iteration:  1200 / 5000 [24%] (Sampling)
## Chain 3 Iteration:  1400 / 5000 [28%] (Sampling)
## Chain 1 Iteration:  1400 / 5000 [28%] (Sampling)
## Chain 4 Iteration:  1400 / 5000 [28%] (Sampling)
## Chain 3 Iteration:  1600 / 5000 [32%] (Sampling)
## Chain 2 Iteration:  1200 / 5000 [24%] (Sampling)
## Chain 4 Iteration:  1600 / 5000 [32%] (Sampling)
## Chain 3 Iteration:  1800 / 5000 [36%] (Sampling)
## Chain 1 Iteration:  1600 / 5000 [32%] (Sampling)
## Chain 2 Iteration:  1400 / 5000 [28%] (Sampling)
## Chain 4 Iteration:  1800 / 5000 [36%] (Sampling)
## Chain 3 Iteration:  2000 / 5000 [40%] (Sampling)
## Chain 1 Iteration:  1800 / 5000 [36%] (Sampling)

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## Chain 4 Iteration: 2000 / 5000 [ 40%] (Sampling)
## Chain 3 Iteration: 2200 / 5000 [ 44%] (Sampling)
## Chain 2 Iteration: 1600 / 5000 [ 32%] (Sampling)
## Chain 3 Iteration: 2400 / 5000 [ 48%] (Sampling)
## Chain 4 Iteration: 2200 / 5000 [ 44%] (Sampling)
## Chain 1 Iteration: 2000 / 5000 [ 40%] (Sampling)
## Chain 3 Iteration: 2600 / 5000 [ 52%] (Sampling)
## Chain 4 Iteration: 2400 / 5000 [ 48%] (Sampling)
## Chain 2 Iteration: 1800 / 5000 [ 36%] (Sampling)
## Chain 1 Iteration: 2200 / 5000 [ 44%] (Sampling)
## Chain 3 Iteration: 2800 / 5000 [ 56%] (Sampling)
## Chain 4 Iteration: 2600 / 5000 [ 52%] (Sampling)
## Chain 3 Iteration: 3000 / 5000 [ 60%] (Sampling)
## Chain 1 Iteration: 2400 / 5000 [ 48%] (Sampling)
## Chain 2 Iteration: 2000 / 5000 [ 40%] (Sampling)
## Chain 4 Iteration: 2800 / 5000 [ 56%] (Sampling)
## Chain 3 Iteration: 3200 / 5000 [ 64%] (Sampling)
## Chain 4 Iteration: 3000 / 5000 [ 60%] (Sampling)
## Chain 1 Iteration: 2600 / 5000 [ 52%] (Sampling)
## Chain 2 Iteration: 2200 / 5000 [ 44%] (Sampling)
## Chain 3 Iteration: 3400 / 5000 [ 68%] (Sampling)
## Chain 4 Iteration: 3200 / 5000 [ 64%] (Sampling)
## Chain 1 Iteration: 2800 / 5000 [ 56%] (Sampling)
## Chain 3 Iteration: 3600 / 5000 [ 72%] (Sampling)
## Chain 4 Iteration: 3400 / 5000 [ 68%] (Sampling)
## Chain 2 Iteration: 2400 / 5000 [ 48%] (Sampling)
## Chain 3 Iteration: 3800 / 5000 [ 76%] (Sampling)
## Chain 4 Iteration: 3600 / 5000 [ 72%] (Sampling)
## Chain 1 Iteration: 3000 / 5000 [ 60%] (Sampling)
## Chain 3 Iteration: 4000 / 5000 [ 80%] (Sampling)
## Chain 2 Iteration: 2600 / 5000 [ 52%] (Sampling)
## Chain 4 Iteration: 3800 / 5000 [ 76%] (Sampling)
## Chain 1 Iteration: 3200 / 5000 [ 64%] (Sampling)
## Chain 3 Iteration: 4200 / 5000 [ 84%] (Sampling)
## Chain 4 Iteration: 4000 / 5000 [ 80%] (Sampling)
## Chain 3 Iteration: 4400 / 5000 [ 88%] (Sampling)
## Chain 1 Iteration: 3400 / 5000 [ 68%] (Sampling)
## Chain 2 Iteration: 2800 / 5000 [ 56%] (Sampling)
## Chain 4 Iteration: 4200 / 5000 [ 84%] (Sampling)
## Chain 3 Iteration: 4600 / 5000 [ 92%] (Sampling)
## Chain 4 Iteration: 4400 / 5000 [ 88%] (Sampling)
## Chain 1 Iteration: 3600 / 5000 [ 72%] (Sampling)
## Chain 3 Iteration: 4800 / 5000 [ 96%] (Sampling)
## Chain 2 Iteration: 3000 / 5000 [ 60%] (Sampling)
## Chain 4 Iteration: 4600 / 5000 [ 92%] (Sampling)
## Chain 3 Iteration: 5000 / 5000 [100%] (Sampling)
## Chain 3 finished in 38.2 seconds.
## Chain 1 Iteration: 3800 / 5000 [ 76%] (Sampling)
## Chain 4 Iteration: 4800 / 5000 [ 96%] (Sampling)
## Chain 2 Iteration: 3200 / 5000 [ 64%] (Sampling)
## Chain 4 Iteration: 5000 / 5000 [100%] (Sampling)
## Chain 1 Iteration: 4000 / 5000 [ 80%] (Sampling)
## Chain 4 finished in 41.2 seconds.
## Chain 2 Iteration: 3400 / 5000 [ 68%] (Sampling)

```

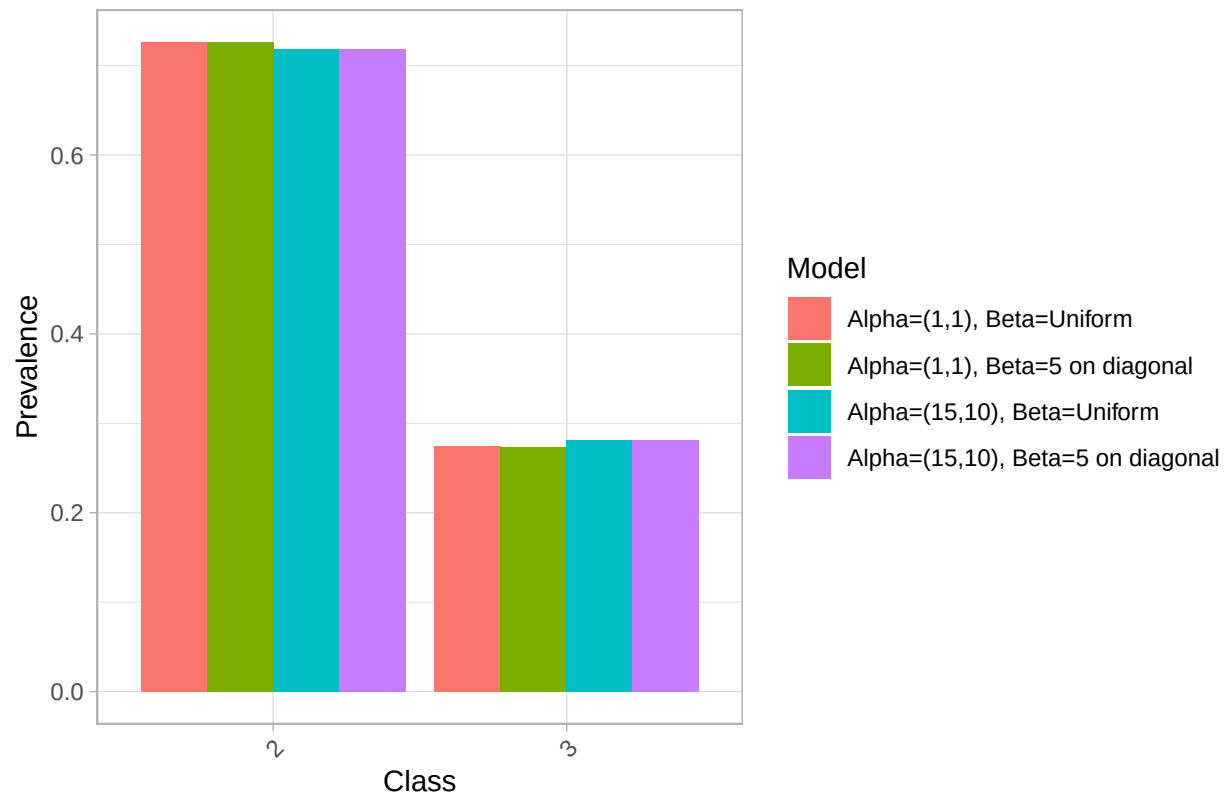


```

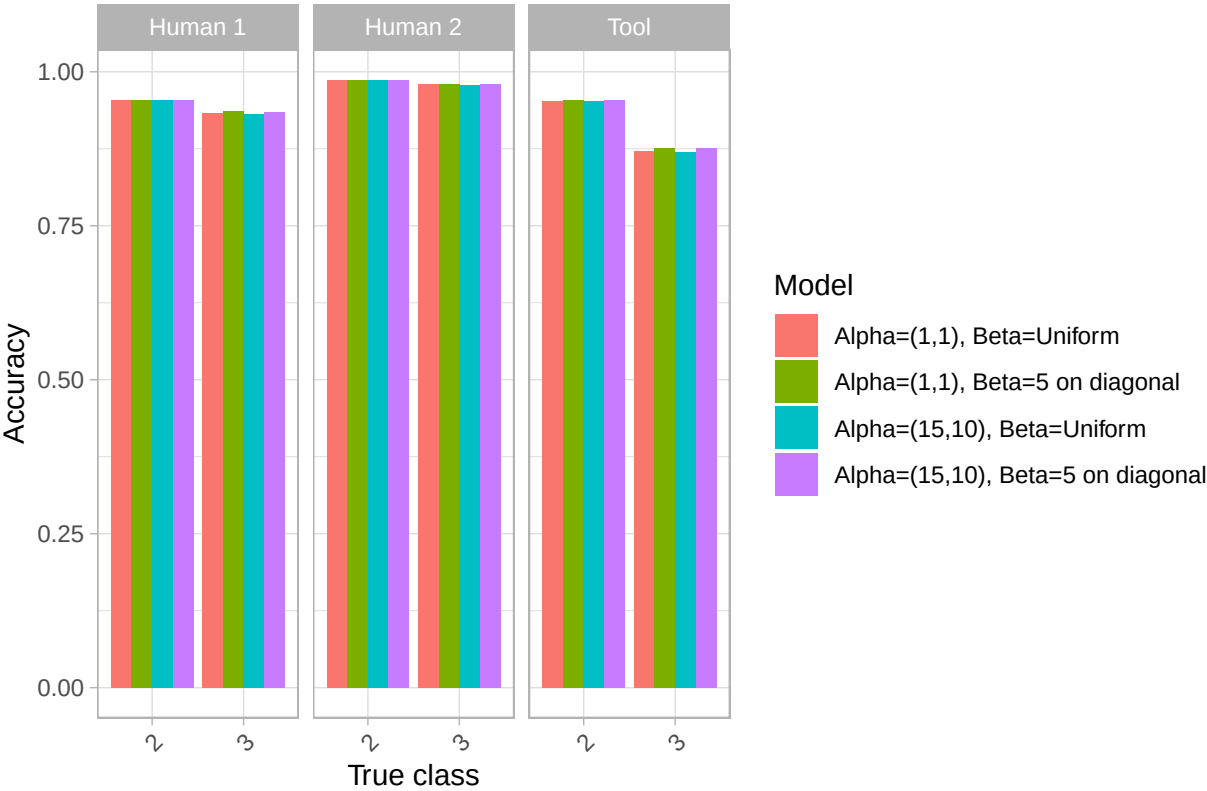
## Chain 1 Iteration: 4200 / 5000 [ 84%] (Sampling)
## Chain 2 Iteration: 3600 / 5000 [ 72%] (Sampling)
## Chain 1 Iteration: 4400 / 5000 [ 88%] (Sampling)
## Chain 2 Iteration: 3800 / 5000 [ 76%] (Sampling)
## Chain 1 Iteration: 4600 / 5000 [ 92%] (Sampling)
## Chain 1 Iteration: 4800 / 5000 [ 96%] (Sampling)
## Chain 2 Iteration: 4000 / 5000 [ 80%] (Sampling)
## Chain 1 Iteration: 5000 / 5000 [100%] (Sampling)
## Chain 1 finished in 52.4 seconds.
## Chain 2 Iteration: 4200 / 5000 [ 84%] (Sampling)
## Chain 2 Iteration: 4400 / 5000 [ 88%] (Sampling)
## Chain 2 Iteration: 4600 / 5000 [ 92%] (Sampling)
## Chain 2 Iteration: 4800 / 5000 [ 96%] (Sampling)
## Chain 2 Iteration: 5000 / 5000 [100%] (Sampling)
## Chain 2 finished in 63.6 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 48.8 seconds.
## Total execution time: 63.7 seconds.
##
## No R-hat values above 1.1
##
## === Analyzing model: alpha11_beta1 ===
##
## === Analyzing model: alpha11_beta5diag ===
##
## === Analyzing model: alpha1510_beta1 ===
##
## === Analyzing model: alpha1510_beta5diag ===

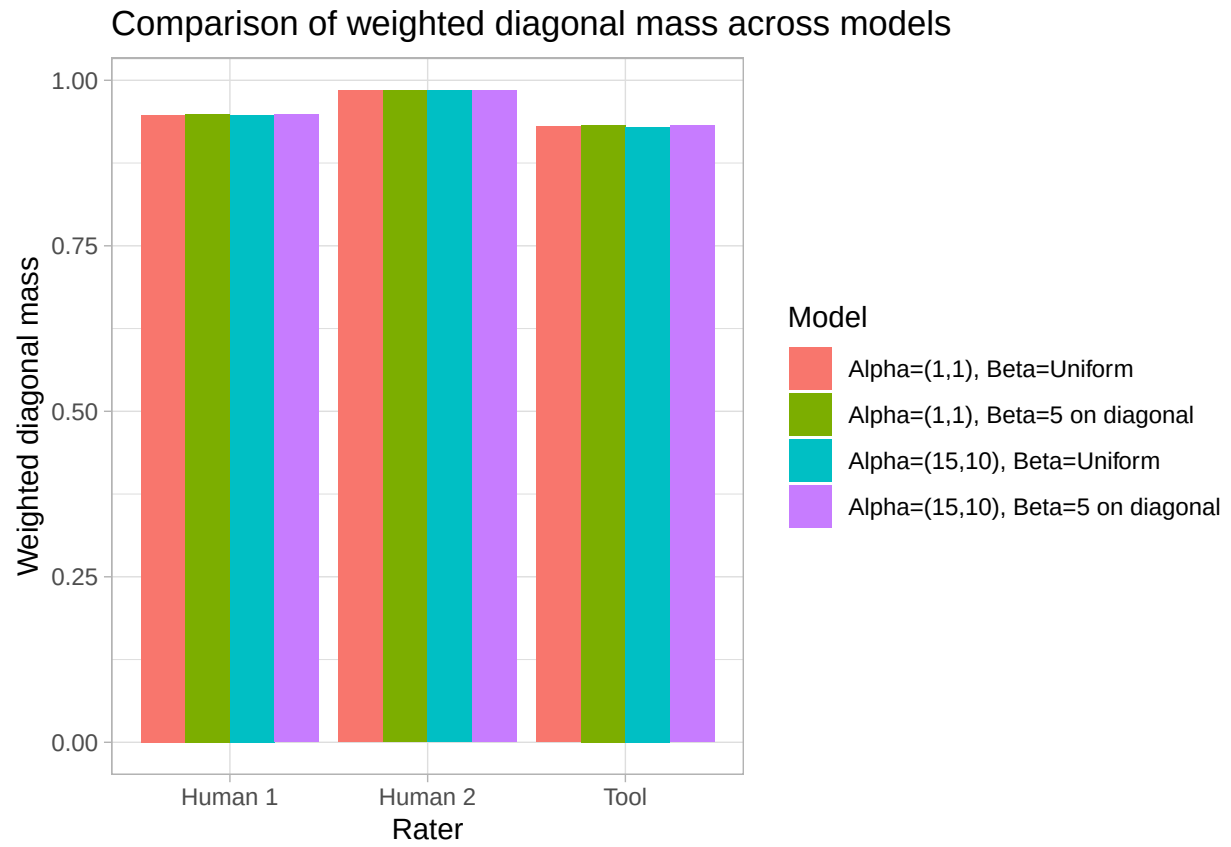
```

Comparison of posterior mean p across models (ISCO categories 2 and 3)

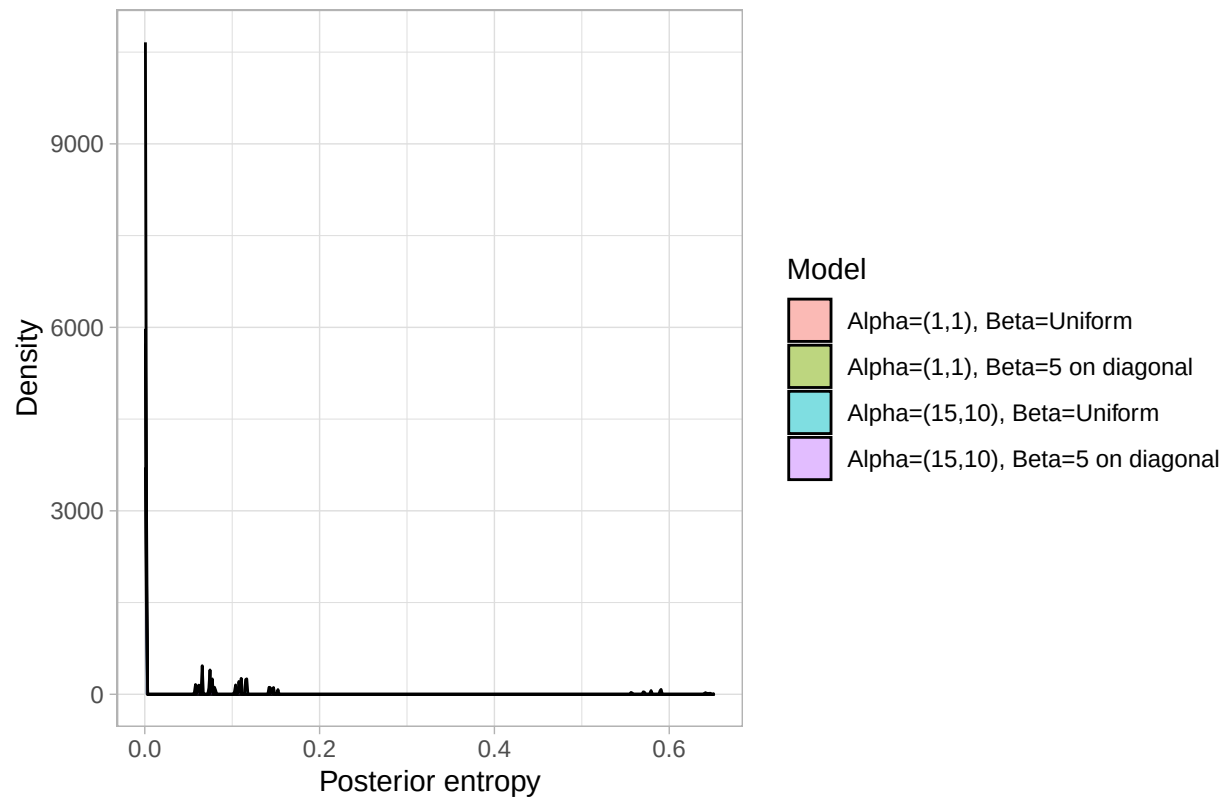


Comparison of diagonal elements (accuracy) across models

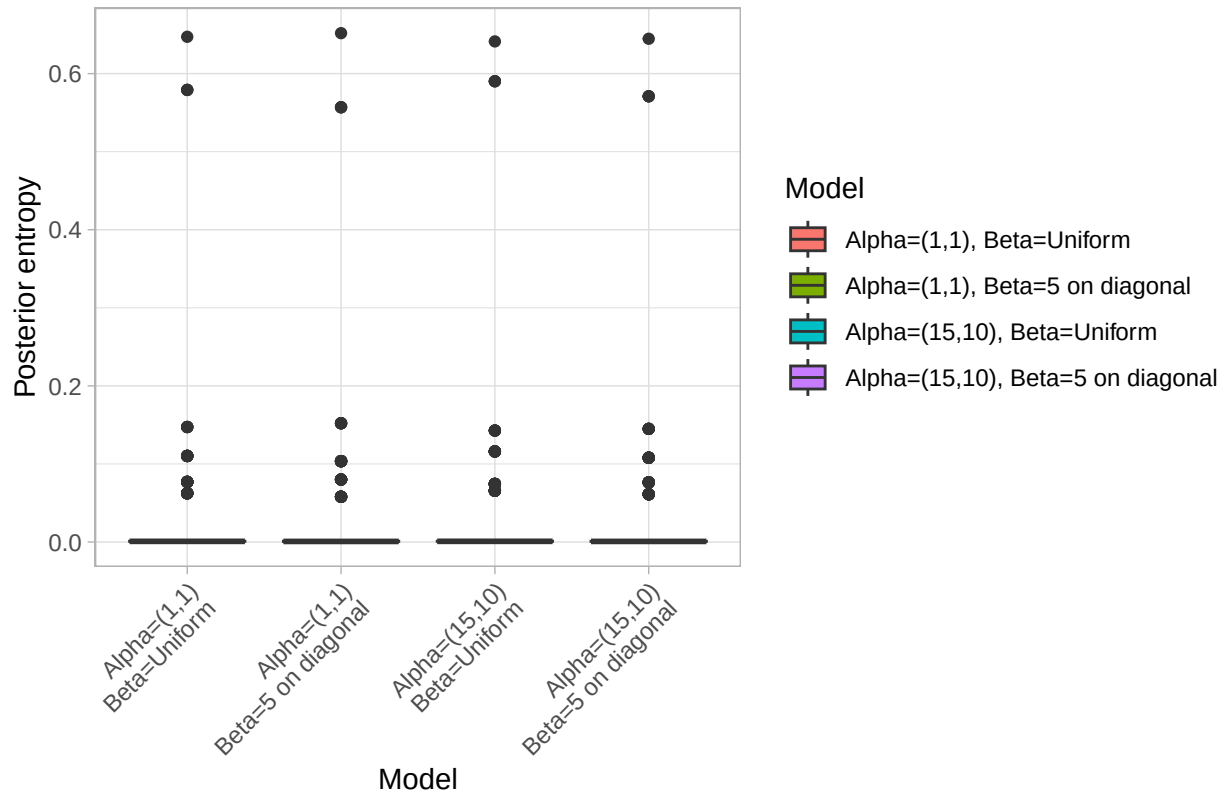




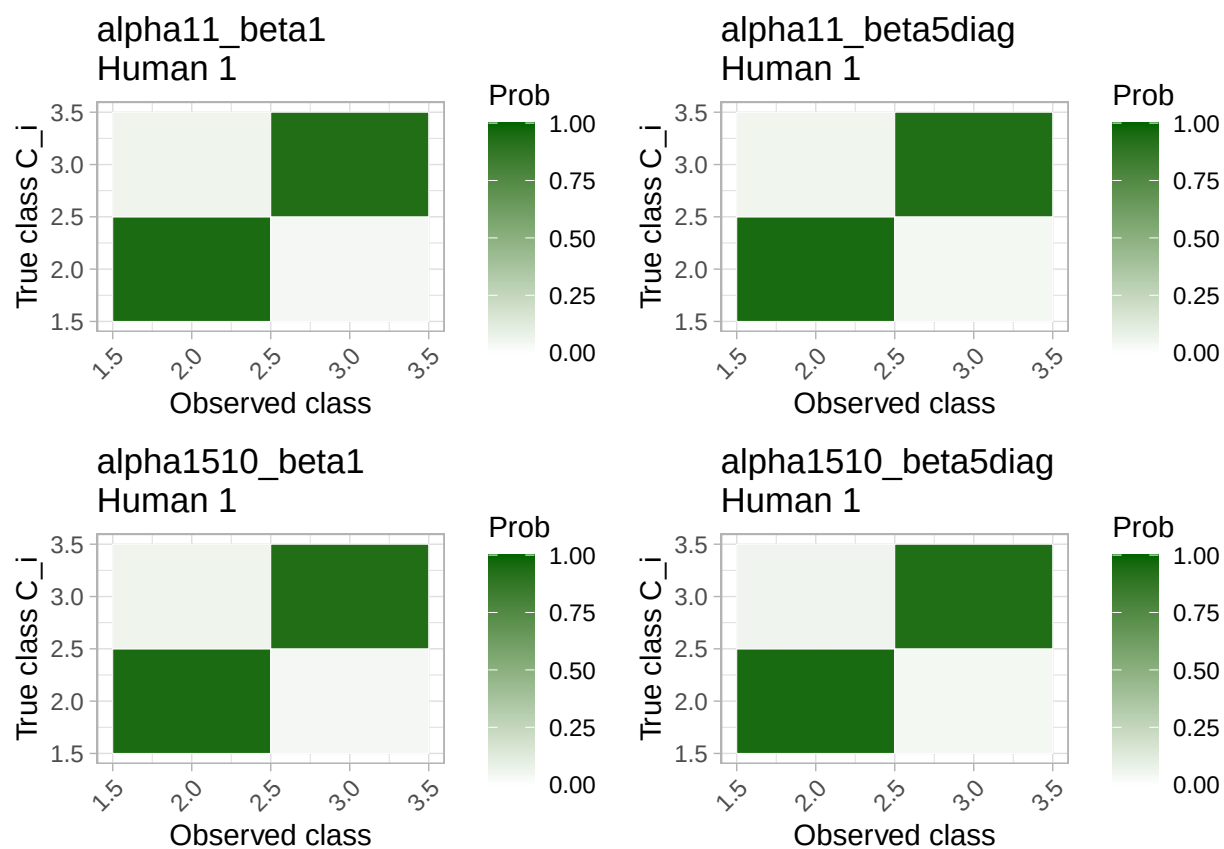
Comparison of item uncertainty (entropy) distributions

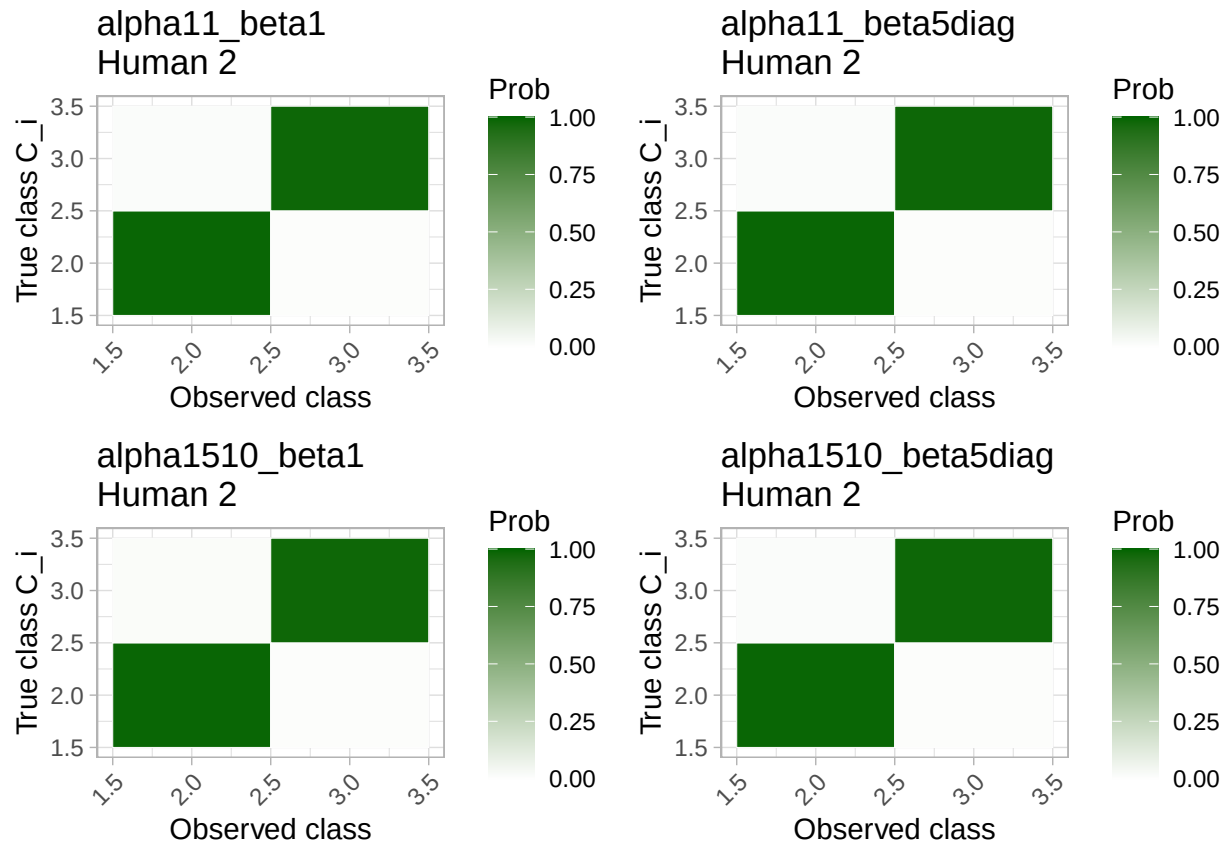


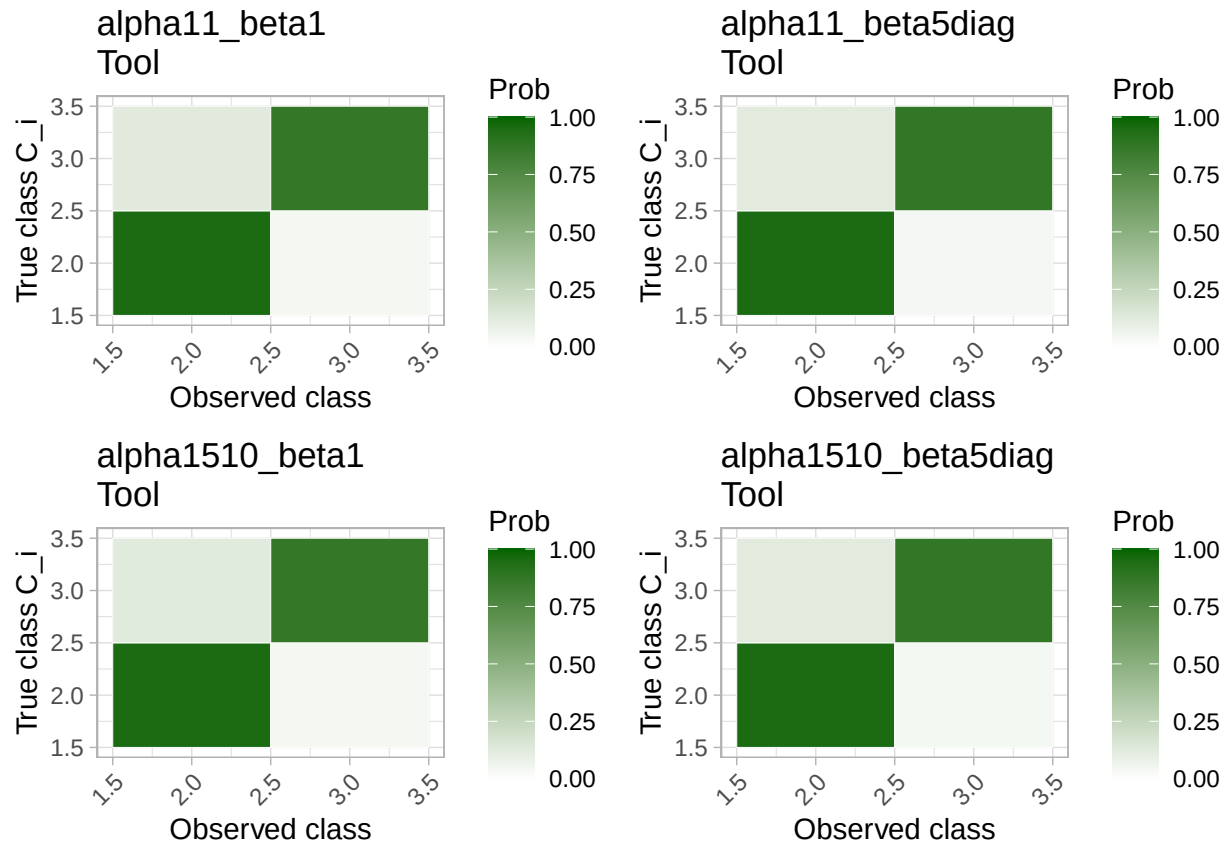
Comparison of item uncertainty (entropy) – boxplot



##	model	pi_class2	pi_class3	h1_weighted_diag	h2_weighted_diag	tool_weighted_diag	
##	pi[1]	alpha11_beta1	0.7257777	0.2742223	0.9474405	0.9846183	0.9302728
##	pi[1]1	alpha11_beta5diag	0.7261432	0.2738568	0.9487450	0.9847980	0.9321658
##	pi[1]2	alpha1510_beta1	0.7185171	0.2814829	0.9471844	0.9845082	0.9294196
##	pi[1]3	alpha1510_beta5diag	0.7188930	0.2811070	0.9483932	0.9848377	0.9316639
##	mean_entropy	median_entropy	mean_max_p	median_max_p			
##	pi[1]	0.01721223	0.0008302909	0.9947175	0.9999205		
##	pi[1]1	0.01678137	0.0007192873	0.9949205	0.9999321		
##	pi[1]2	0.01744279	0.0008766941	0.9946145	0.9999155		
##	pi[1]3	0.01691561	0.0007607126	0.9948428	0.9999278		







```
##
## === Comparison complete ===
## All comparison plots saved to: C:/Users/jeroe/Uni_C/noisy_measurement/img/model_results/isco/alpha_b
```